

ARC Generalization Lab — One-Page Concept Summary

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1. Central Claim

High performance on a public benchmark does not, by itself, establish generalization to novel tasks. A public-versus-fresh, distribution-matched comparison provides **diagnostic evidence** about a possible generalization gap—but does not prove memorization. The ARC Generalization Lab operationalises this claim as an interactive research instrument: the same solver evaluates both public ARC-AGI-1 tasks (which appear in web-scale training corpora) and freshly generated, distribution-matched tasks (which cannot), letting the learner observe and interpret the resulting performance gap.

2. Why This Matters

Bordes et al. (2024) demonstrated empirically that ARC-AGI-1 evaluation tasks appear verbatim in Common Crawl and The Pile. Any model pre-trained on these corpora may score higher than its genuine rule-induction ability warrants. Without a fresh, distribution-matched control set, static benchmark scores conflate prior familiarity with general reasoning capability.

3. Competitive Context & Architectural Comparison

We situate BDH-CQ against two representative approaches on dimensions that genuinely differentiate them:

Dimension	OpenAI o3	TTT (Akyürek 2024)	BDH-CQ (Engdahl 2026)
Public ARC-AGI accuracy	75.7% pass@3	~53% (TTT + 8B)	29.5% pass@2
Test-time param updates	Undisclosed (CoT)	Yes (gradient-based)	None (W fixed)
Inference cost per task	~\$17–33 (est.)	Moderate (fine-tune)	\$0.0007
Parameters	Undisclosed (frontier)	8B (Llama 3)	150M
Fresh-task evaluation	Not published	Not published	Dev-reported only

4. Mechanism: Optimization vs. Contextual Adaptation

Optimization-based approaches (TTT, HRM/TRM) update model weights on each test instance via gradient descent. BDH-CQ takes the opposite path: weights remain frozen at inference time. Instead, each demonstration updates an evolving recurrent associative memory state $S(t)$ through forward-pass binding operations. The model accumulates task-specific context without backpropagation, enabling inference at \$0.0007 per task with 150M parameters.

5. Strengths and Limitations

Strengths: BDH-CQ achieves the lowest reported per-task inference cost on ARC-AGI-1 by orders of magnitude. Its parameter-frozen inference eliminates catastrophic forgetting risk during evaluation and requires no per-task fine-tuning infrastructure. The 150M parameter footprint makes it deployable on consumer hardware.

Where BDH-CQ performs worse: Raw accuracy (29.5% pass@2) is substantially below frontier systems—OpenAI o3 reaches 75.7% and Greenblatt's GPT-4o pipeline achieves 72% pass@2. BDH-CQ has not been independently evaluated on freshly generated tasks; all published figures are developer-reported on the public evaluation set. The finite-capacity associative state may saturate on tasks requiring many distinct rules or long demonstration sequences.

6. Maturity Assessment

BDH-CQ is at an **early evaluation stage**. Developer-reported results on public ARC-AGI-1 exist (Engdahl et al., 2026), but no independent external reproduction has been published as of September 2026. The architecture has not been stress-tested on out-of-distribution abstract reasoning benchmarks beyond ARC. Our lab cites all BDH-CQ figures as PUBLISHED RESULT and does not claim independent verification.

7. Evidence Classification

Every datum in the artifact is labelled: **PUBLISHED RESULT** (cited from literature), **MEASURED** (computed live by local solver), **DEMO DATA** (hand-crafted tutorials), or **TOY / DIDACTIC** (simplified educational model). No result is presented without provenance.

8. Where to Learn More & Primary References

Next Steps: Explore the live interactive artifact at frontend-navy-gamma-81.vercel.app to run the 60-second diagnostic journey, inspect task-generation code in our repository (github.com/Astha167/ARC_Generalization_Lab), and consult primary sources:

- Engdahl et al. (2026). *BDH-CQ: Introducing In-Context Learning with Recurrent Latent Reasoning*. arXiv:2608.09888.
- Bordes et al. (2024). *An In-Depth Look at Gemini's ARC-AGI Capabilities and Contamination*. arXiv:2407.00645.
- Akyürek et al. (2024). *The Surprising Effectiveness of Test-Time Training for Abstract Reasoning*. arXiv:2411.07279.
- Chollet (2019). *On the Measure of Intelligence*. arXiv:1911.01547.